NMC 2014 A-LEVEL PAPER 1 QUESTIONS

SECTION A

Qn1. Let $2007! + 2008! + 2009! = (a!)(b^2)$, where a and b are positive integers. Without using a calculator, find the pair (a, b).

Qn2. The function f has the following properties: $f(1) = 1$ and $f(n) = n + f(n-1)$ for all integers $n \geq 2$. Compute $f(2014)$.

Qn3. Let S be a set of odd numbers $85, 99, 117$ and 121. Which of these numbers could be the discriminant D of the quadratic equation $ax^2 + bx + c = 0$, where $a > 0$, a, b, c are integers and $D = b^2 - 4ac$.

Qn4. In a mathematics class, students grades are based on tests only, all of which have equal weight. When Eva got a grade of 98 on a certain test, it raised her average by 1 point. But when she got a grade of 70 on the next test, her average then dropped by 2 points. Including these two tests, how many tests did she take?

Qn5. The sum of the 3−digit numbers $35x$ and $4y7$ is divisible by 36. Find all possible ordered pairs (x, y).

Qn6. Jack, John and James are identical triplets. It is impossible to distinguish them by appearance. Jack and John always tell the truth, but James always lies i.e everything he says is false. You know that the triplets are between 20 and 30 years old, 20 and 30 included. One day you meet two of the triplets and ask them how old they are. A says "We are between 20 and 29 years old, 20 and 29 included". B makes the following statements: "We are between 21 and 30 years old, 21 and 30 included" and "One of us present is lying". How old are they?

Qn7. If $m^2 + n^2 = 12$ and $m^4 + n^4 = 136$, determine all possible integral values of m and n.

Qn8. The solutions of the equation $x^2 + px + q = 0$ are the cubes of the solutions of the equation $x^2 + mx + n = 0$. Find the relationship between p, m and n. Hence, find p if $nm = 4$ and $m = 3$.

Qn9. Let $a = -\sqrt{3} + \sqrt{5} + \sqrt{7}$, $b = \sqrt{3} - \sqrt{5} + \sqrt{7}$ and $c = \sqrt{3} + \sqrt{5} - \sqrt{7}$. Evaluate

$$\frac{a^4}{(a-b)(a-c)} + \frac{b^4}{(b-a)(b-c)} + \frac{c^4}{(c-b)(c-a)}.$$

Qn10. Points A, B, C, and D lie in a plane, with A, B, and C in order in a straight line. The angles between some of these points, measured in degrees, satisfy $BDC = 78$, $DBC > DCB$, $ABD = 4x + y$ and $DCB = x + y$, for some numbers x and y. How many positive integer values can y take on, satisfying all these conditions?

SECTION B

Qn11. (a) What is the value of $\frac{2^{2014}+2^{2012}}{2^{2014}-2^{2012}}$.

(b) The sums of three whole numbers taken in pairs are 12, 17, and 19. What is the middle number?

Qn12. Let a, b and c be distinct positive integers that form a Geometric Progression and $b-a$ be a prime number between 0 and 10 such that

$$\log_4 a + \log_4 b + \log_4 c = 3.$$

Find the highest value of $a+b+c$.

Qn13. A bus, an hour after starting meets an accident which detains it for half an hour, after which it proceeds at $\frac{3}{4}$ of its former speed and arrives $3\frac{1}{2}$ hours late. Had the accident happened $90km$ further along the road, it would have arrived only 3 hours late. How long is the trip.

Qn14. Find all possible positive integers (a, b) that satisfy the equation

$$(a+1) + (b+1) + ab = 2014.$$

Qn15. One of the sides of a triangle PQR is divided into segments of 6 and 8 units by the point of tangency of the inscribed circle. If the radius of the circle is 4 units, find the length of the shortest side of the triangle.

NMC 2014 A-LEVEL PAPER 1 SOLUTIONS
SECTION A

1.

$$\begin{aligned} 2007! + 2008! + 2009! &= 2007! + 2008(2007!) + 2009(2008)(2007!). \\ &= 2007!(1 + 2008 + 2009(2008)). \\ &= 2007!(2009)(1 + 2008) \\ &= (2007!)(2009)^2. \end{aligned}$$

Comparing with $(a!)(b^2)$ we have $(a, b) = (2007, 2009)$

2.

$$\begin{aligned} f(n) &= n + f(n-1) \\ f(2014) &= 2014 + f(2013) \\ f(2013) &= 2013 + f(2012) \\ f(2012) &= 2012 + f(2011) \end{aligned}$$

$$\vdots$$

$$\begin{aligned} f(2) &= 2 + f(1) \\ \text{Therefore } f(2014) &= 2014 + 2013 + \cdots 1 \end{aligned}$$

$$S_{2014} = \frac{n(n+1)}{2} = \frac{2014(2014+1)}{2} = \frac{2014(2015)}{2} = 2029105.$$

3. The discriminant of $ax^2 + bx + c = 0$ is $D = b^2 - 4ac$. Since all of the choices are odd numbers, and $4ac$ is even, then b^2 and hence b, must be odd. Let $b = 2n + 1$ where n is an integer. Then

$$b^2 - 4ac = (2n+1)^2 - 4ac = 4n^2 + 4n + 1 - 4ac = 4n^2 + 4n - 4ac + 1 = 4(n^2 + n - ac) + 1.$$

Therefore, the discriminant D must be one more than a multiple of 4. Of the choices, only 99 does not meet this requirement. The rest of the numbers could be a discriminant.

4. Suppose before scoring 98 and 70, the student's average grade on her first N tests was M. Then

$$\frac{NM + 98}{N+1} = M + 1 \quad \text{and} \quad \frac{NM + 98 + 70}{N+2} = M + 1 - 2$$

The first equation simplifies to $M + N = 97$ and the second to $2M - N = 170$. Thus solving these two equations simultaneously gives $M = 89$ and $N = 8$. Including the two tests, she took 10 tests.

5. We have $350+x+407+10y = 36j$ for some j. Since $757 = 36(21)+1$, we have $10y+x = 36k-1$ for some k. Since $0 \leq 10y + x \leq 99$, we have $10y + x = 35$ or 71. Now suppose $10y + x = 35$, then $x = 5$ and $y = 3$. Also, if $10y + x = 71$, then $x = 1$ and $y = 7$. Therefore the possible ordered pairs (x, y) are $(5, 3)$ and $(1, 7)$.

6. It follows from the fact that B says "One of us present is lying", that he must be telling the truth. Indeed, suppose that B were lying. Then his utterance "One of us present is lying" would be a false statement. Hence both present would be telling the truth, which would contradict the assumption that A is lying. Since we now know that what B is saying is true, we can deduce that A must be lying, hence that the triplets are not between 20 and 29 years old, and hence that they must be 30 years old.

7. **We have that** $m^2 + n^2 = 12$ **and** $m^4 + n^4 = 136$**.**

   From $(n^2 + m^2)^2 = m^4 + n^4 + 2n^2m^2$, we have

   $\frac{12^2}{2}$

8. Let x_1 and x_2 be the roots of $x^2 + mx + n = 0$. This implies x_1^3 and x_2^3 are the roots of $x^2 + px + q = 0$.

   Thus $x_1 + x_2 = -m$ and $x_1x_2 = n$; $\quad x_1^3 + x_2^3 = -p$ and $x_1^3x_2^3 = q$.

   From $(x_1 + x_2)^3 = (x_1^3 + x_2^3) + 3x_1x_2(x_1 + x_2) = -p + 3n(-m)$.

   Therefore $\quad m^3 = p + 3nm$.

   We thus have $p = m^3 - 3nm = 3^3 - 3(4) = 27 - 12 = 15$.

9. Let $a = -\sqrt{3} + \sqrt{5} + \sqrt{7}$, $b = \sqrt{3} - \sqrt{5} + \sqrt{7}$ and $c = \sqrt{3} + \sqrt{5} - \sqrt{7}$. Evaluate

$$\frac{a^4}{(a-b)(a-c)} + \frac{b^4}{(b-a)(b-c)} + \frac{c^4}{(c-b)(c-a)}.$$

10.

$$\begin{aligned} \alpha + \beta &= 102. && (1)\\ 108\alpha &= 4x + y. && (2)\\ x + y &= \beta. && (3) \end{aligned}$$

Substituting Equations (2) and (3) into Equation (1) we get

$$\begin{aligned} 180 - 4x - y + x + y &= 102.\\ 78 - 3x &= 0.\\ x &= 26. \end{aligned}$$

Then $76y = \alpha > \beta = 26 + y$, so, $y < 25$. Therefore the values of y range from 1 to 24.

SECTION B

11. (a) Factoring 2^{2012} from each of the terms and simplifying gives

$$\frac{2^{2014}+2^{2012}}{2^{2014}-2^{2012}} = \frac{2^{2012}(2^2+1)}{2^{2012}(2^2-1)} = \frac{4+1}{4-1} = \frac{5}{3}.$$

(b) Let the three whole numbers be $a < b < c$. The set of sums of pairs of these numbers is $(a+b, a+c, b+c) = (12, 17, 19)$. Thus

$$2(a+b+c) = (a+b)+(a+c)+(b+c) = 12+17+19 = 48.$$

Dividing by 2 gives

$$a+b+c = 24.$$

If follows that $(a,b,c) = (24-19, 24-17, 24-12) = (5,7,12)$. Therefore the middle number is 7.

12. If a, b, c form a Geometric progression, then $\frac{b}{a} = \frac{c}{b} \Longrightarrow b^2 = ac$. From

$$\begin{aligned} \log_4 a + \log_4 b + \log_4 c &= 3. \\ \log_4(abc) &= 3 \\ abc &= 4^3. \end{aligned}$$

From $b^2 = ac$ and $abc = 4^3$, we have that $b^3 = 4^3 \Longrightarrow b = 4$. Now since $b-a$ is a prime number between 0 and 10, then

$$b-a = \{2,3,5,7\}.$$

$$4-a = \{2,3,5,7\}.$$

The only integers satisfying this condition are $a = 2$ and $a = 1$.

However, if $a = 1$, then $abc = 4^3 \Longrightarrow c = 4^2 = 16$.

Thus $(a,b,c) = (1,4,6)$. This is a geometric progression whose sum is 21.

Now if $a = 2$, then $abc = 4^3 \Longrightarrow c = 8$.

Thus $(a,b,c) = (2,4,8)$. This is also a geometric progression whose sum is 14. Therefore the highest value of $a+b+c$ is 21.

13. Let X be the distance from the point of the accident to the end of the trip and R the former speed of the bus. Then the normal time for the trip in hours is $\frac{X}{R}+1$.

Considering the time for each trip, we have

$$1+\frac{1}{2}+\frac{X}{\frac{3}{4}R} = \frac{X}{R}+1+3\frac{1}{2}$$

and

$$1 + \frac{90}{R} + \frac{1}{2} + \frac{X - 90}{\frac{3}{4}R} = \frac{X + R}{R} + 3$$

Simplifying these two equations gives

$$\frac{X}{R} = 9 \Longrightarrow X = 9R \tag{4}$$

and

$$-180 + 2X = 15R. \tag{5}$$

Solving equations (7) and (8) gives

$$\begin{aligned} -180 + 18R &= 15R. \\ -180 &= -3R. \\ R &= 60. \end{aligned}$$

Thus $X = 9R = 60 \times 9 = 540$. Therefore the trip is $540 + 60 = 600$km.

14.

$$\begin{aligned} (a+1) + (b+1) + ab &= 2014 \\ (a+1) + (b+1) + ab &= 2014. \\ a + b + ab + 2 &= 2014. \\ (a+1)(b+1) &= 2013. \\ \text{but } 2013 &= 3 \times 11 \times 61 \text{ prime factorisation.} \\ (a+1)(b+1) &= 3 \times 11 \times 61. \end{aligned}$$

The possibilities are

(a) $a + 1 = 1, b + 1 = 2013$ which means that $a = 0$. This is not possible.

(b) $a + 1 = 3$ and $b + 1 = 11 \times 61$. Hence $(a, b) = (2, 670)$.

(c) $a + 1 = 11$ and $b + 1 = 3 \times 61$. Hence $(a, b) = (10, 182)$.

(d) $a + 1 = 61$ and $b + 1 = 3 \times 11$. Hence $(a, b) = (60, 32)$.

(e) $a + 1 = 3 \times 11$ and $b + 1 = 61$. Hence $(a, b) = (32, 60)$.

(f) $a + 1 = 3 \times 61$ and $b + 1 = 11$. Hence $(a, b) = (182, 10)$.

(g) $a + 1 = 61 \times 11$ and $b + 1 = 3$. Hence $(a, b) = (670, 2)$.

Therefore the ordered pairs are $(a, b) = (2, 670), (10, 182), (60, 32), (32, 60), (182, 10), (670, 2)$.

15. Denote the sides of a triangle by a, b, c where $a = 8+6 = 14$, $b = 8+x$ and $c = x+6$. Therefore

$$\begin{aligned} 2s &= a+b+c \\ &= 14+8+x+x+6. \\ &= 2x+28. \\ s &= x+14. \end{aligned}$$

On the other hand, by Heron's Formula,

$$\begin{aligned} Area\ of\ \triangle ABC &= \sqrt{s(s-a)(s-b)(s-c)} \\ &= \sqrt{(x+14)(x+14-14)(x+14-8-x)(x+14-x-6)}. \\ &= \sqrt{48x(x+14)}. \end{aligned}$$

From the fact that the area of a triangle is half the product of the perimeter and the radius of the inscribed circle, we have that

$$\begin{aligned} 4(x+14) &= \sqrt{48x(x+14)} \\ (x+14)^2 &= 3x(x+14). \\ x+14 &= 3x. \\ 2x &= 14. \\ x &= 7 \end{aligned}$$

Therefore the sides of the triangle are $c = 7+6 = 13$, $a = 6+8 = 14$ and $b = 8+7 = 15$.

Thus the shortest side is 13 units.

Alternatively
This could as well be done by construction method.